THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
KILIMANJARO REGIONAL COMMISSIONER'S OFFICE

FORM FOUR POST MOCK EXAMINATION

032/2 CHEMISTRY 2A (ACTUAL PRACTICAL)

2:30 Hours **Wednesday 24th September, 2025**

INSTRUCTIONS

1. This paper consists of **two (2)** questions. Answer **all** the questions
2. Each question carries **twenty-five (25)** marks
3. Qualitative Analysis Guide sheet authorized by NECTA and non-programmable may be used
4. Cellular phones are **not** allowed in examination room
5. Write your **Examination number** on every page of your answer booklet(s)
6. You may use the following constant.
   Atomic masses H=1, C=12, Na=23, Cl=35.5
   1 litre =1 dm^3=1000 cm^3

1. You are provided with;

   1.1. Solution **F** containing 3.65 g/$dm^3$of HW acid solution per cubic decimeter of solution.

   1.2. Solution **G** containing 2.0g of sodium hydroxide in 500cm^3.

   1.3. Methyl orange indicator.

Procedure:

Put solution **F** into a burette. Pipette 25cm^3 or 20cm^3 of solution **G** into a titration flask. Add two drops of Methyl orange indicator. Titrate the solution **F** against solution **G** until a colour change is observed. Repeat the procedure to obtain three more readings and record your results in tabular form.

Questions

(a) How much volume of the acid required to neutralize completely 20cm^3 or 25cm^3 of base?

(b) The colour change during titration was from…………….... to ……………………

(c) Write balanced chemical equation for this reaction

(d) Calculate the molarity of the **G** and that of **F**

(e) Find the molar mass of HW and identify element W

2. Sample **J** is a simple salt containing **one cation** and **one anion.** Carefully carry out all the experiments described in the Table 2. Record all your observations and appropriate inferences to identify the cation and anion present in sample **J.**

Table 2

S/n	Experiments	Observation	Inference
(a)	Appearance of the solid sample **J**		
(b)	Put a spatulaful of sample **J** in a test tube and heat.		
(c)	Add dilute hydrochloric acid to the solid sample **J**		
(d)	Perform flame test on the sample.		
(e)	Dissolve the sample in water. Divide the resulting solution into two portions:		
	(i) Add sodium hydroxide solution to the first portion.		
	(ii) Add few drops of magnesium sulphate solution to the second portion, then warm.		

Conclusion

(i) The cation is ……………………………

(ii) The anion is …………………………….

(iii) The sample **J** is_…………………………

(iv) Write chemical equation (with state symbols) to show reaction which took place in the experiment 2(c)

(v) Describe the confirmatory test for cation